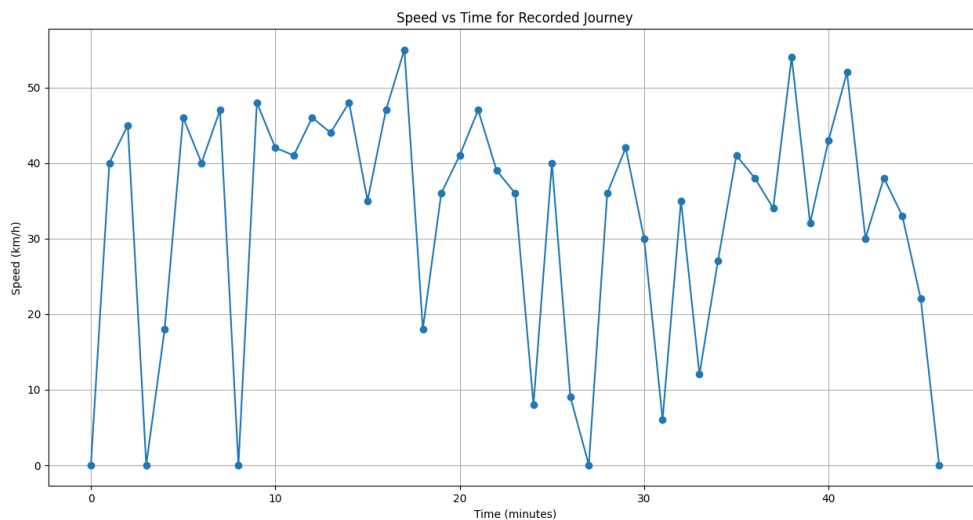


Time (in minutes)

time = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46]

Speed (in km/h)(per 1 minute)

speed = [0, 40, 45, 0, 18, 46, 40, 47, 0, 48, 42, 41, 46, 44, 48, 35, 47, 55, 18, 36, 41, 47, 39, 36, 8, 40, 9, 0, 36, 42, 30, 6, 35, 12, 27, 41, 38, 34, 54, 32, 43, 52, 30, 38, 33, 22, 0]



```
Percentage_Error_Distance.py > ...
1 # Speed data (km/h)
2 speed = [0, 40, 45, 0, 18, 46, 40, 47, 0, 48, 42, 41, 46, 44, 48, 35, 47, 55, 18, 36, 41]
3
4 # Time interval = 1 minute = 1/60 hour
5 distance_each_min = []
6
7 for s in speed:
8     distance_each_min.append(s * (1/60))
9
10 # Total calculated distance
11 calculated_distance = sum(distance_each_min)
12
13 # Actual distance from Google Maps (ENTER YOUR VALUE HERE)
14 actual_distance = 10.0 # km
15
16 # Percentage error
17 percentage_error = abs(actual_distance - calculated_distance) / actual_distance * 100
18
19 print("Calculated distance (km):", round(calculated_distance, 2))
20 print("Actual distance (km):", actual_distance)
21 print("Percentage error (%)ate", round(percentage_error, 2))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS E:\Mtech(Automotive_Technology_2025-2027)\Sem_2\2_Automotive_Intelligence\Assignments\1_MAP> & "C:/Python 3.9/python.exe" "e:/Mtech(Automotive_Technology_2025-2027)/Sem_2/2_Automotive_Intelligence/Assignments/1_MAP/Percentage_Error_Distance.py"
Calculated distance (km): 12.28
Actual distance (km): 10.0
Percentage error (%): 22.83
```

